

2024
Higher School Certificate
Trial Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student name and/or number at the top of every page

Total marks – 100

Section I – 10 marks (pages 3 - 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 - 22)

- Attempt Questions 11 – 32
- Allow about 2 hours and 45 minutes for this section

This paper MUST NOT be removed from the examination room.

STUDENT NAME/NUMBER.....

STUDENT NAME/NUMBER.....

Section I

10 Marks

Attempt Questions 1-10.

Allow about 15 minutes for this section.

Select the alternative A, B, C, D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Section I**10 Marks****Attempt Questions 1-10.****Allow about 15 minutes for this section.****Use the multiple-choice answer sheet for questions 1-10.**

1. If $f(x) = x(x-1)$ how many different solutions are there to the equation $f(x^2) = 0$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

2. ABC is a triangle in which $\tan A = -\frac{4}{3}$. What is the value of $\sin A$?

- (A) $-\frac{4}{5}$
- (B) $-\frac{3}{5}$
- (C) $\frac{3}{5}$
- (D) $\frac{4}{5}$

3. If $f(x) = \log_e x$ and $g(x) = x^2 + 1$ what is the range of the composite function $f(g(x))$?

- (A) $[0, \infty)$
- (B) $[1, \infty)$
- (C) $(0, \infty)$
- (D) $(-\infty, \infty)$

4. A and B are two independent events such that $P(\bar{A} \cap B) = 0.4$ and $P(B) = 0.5$. What is the value of $P(A)$?

- (A) 0.1
- (B) 0.2
- (C) 0.3
- (D) 0.4

5. If $y = \sqrt{x^2 + 4}$ what is an expression for $\frac{dy}{dx}$?

- (A) $\frac{dy}{dx} = \sqrt{2x}$
- (B) $\frac{dy}{dx} = \frac{x}{2\sqrt{x^2 + 4}}$
- (C) $\frac{dy}{dx} = \frac{2x}{\sqrt{x^2 + 4}}$
- (D) $\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 4}}$

6. The function $f(x) = e^x$ is transformed by a horizontal translation of $\log_e 2$ units in the negative x direction followed by a horizontal dilation of scale factor 2. What is the new function $g(x)$?

- (A) $g(x) = 2e^{\frac{x}{2}}$
- (B) $g(x) = \frac{1}{2} e^{\frac{x}{2}}$
- (C) $g(x) = \frac{1}{2} e^{2x}$
- (D) $g(x) = 2e^{2x}$

7. Ozzie scored 70 on a test in which the marks were normally distributed with mean 50 and standard deviation 15. What new mark would Ozzie be given if the marks were adjusted to have mean 60 and standard deviation 12 ?

- (A) 74
- (B) 76
- (C) 78
- (D) 80

8. What is an expression for $\int \frac{2x-1}{2x+1} dx$?

- (A) $\frac{x^2 - x}{x^2 + x} + c$
- (B) $x - \frac{1}{2} \log_e (2x + 1) + c$
- (C) $x - \log_e (2x + 1) + c$
- (D) $x - 2 \log_e (2x + 1) + c$

9. An investment of \$4 000 earns interest at a rate of $r\%$ p.a. over a period of 1 year. How much more interest would the investment earn if interest is compounded half-yearly rather than yearly?

- (A) $\$10r^2$
- (B) $\$r^2$
- (C) $\$\frac{1}{10}r^2$
- (D) $\$\frac{1}{100}r^2$

10. What is the limiting sum of the geometric series $\log_e 2 - (\log_e 2)^2 + (\log_e 2)^3 - (\log_e 2)^4 + \dots$?

- (A) $\log_{\frac{e}{2}} 2$
- (B) $\log_{\frac{2}{e}} 2$
- (C) $\log_{2+e} 2$
- (D) $\log_{2e} 2$

Section II**90 Marks****Attempt Questions 11 – 32.****Allow about 2 hours and 45 minutes for this section.****Instructions**

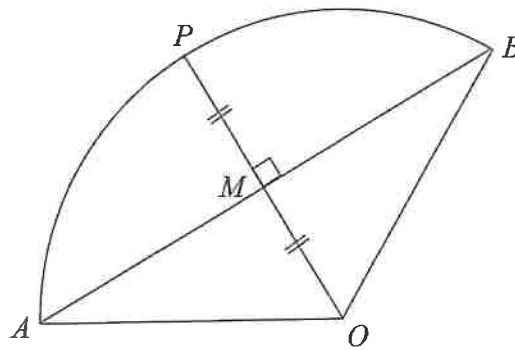
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. If extra space is required, use the lined pages at the end of this section, carefully labelling the additional work with the appropriate question number and part.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)Solve the equation $|2x - 1| = 3$.

2

Question 12 (2 marks)Find the equation of the line through the origin that is perpendicular to the line $\frac{x}{2} + \frac{y}{3} = 1$.

2

Question 13 (3 marks)

In the diagram APB is an arc of a circle with centre O such that AB bisects OP at right angles at M . The length of OP is 12 cm.

(a) Show that $\angle AOB = \frac{2\pi}{3}$.

2

(b) Find in simplest exact form the area of sector AOB .

1

Question 14 (3 marks)

A and B are two events such that $P(A) = 0.3$, $P(B) = 0.2$ and $P(A|B) = 0.5$.

(a) Show that $P(A \cup B) = 0.4$.

2

(b) Find in simplest exact form $P(\text{neither } A \text{ nor } B)$.

1

Marks**Question 15 (3 marks)**Solve the equation $2\log_2(x-6) = \log_2 x$.**3**

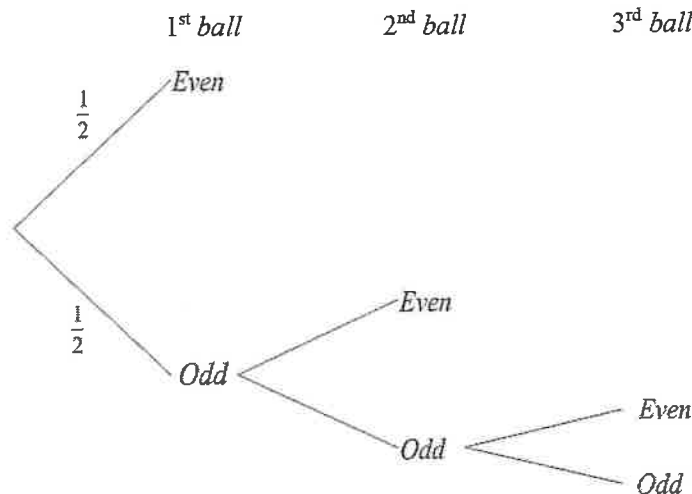
Question 16 (3 marks)Show that $\log_e(1+\sin x) - \log_e(1-\sin x) \equiv 2\log_e(\sec x + \tan x)$.**3**

Question 17 (4 marks)

A bag contains six balls numbered 1, 2, 3, 4, 5, 6 respectively. A ball is chosen at random from the bag. If this first number chosen is even, no further balls are chosen. If the first number chosen is odd, it is put aside and another ball is chosen at random from the remaining balls in the bag. If this second number chosen is even, no further balls are chosen. If the second number chosen is odd, it is put aside and another ball is chosen at random from the remaining balls in the bag. Let the random variable X denote the number of odd numbers chosen.

(a) Complete the probability tree diagram below.

1



(b) Complete the probability distribution for the random variable X in the table below

3

x	0	1	2	3
$p(x)$				

and hence find in simplest exact form the probability that the majority of the numbers chosen are odd numbers.

.....

.....

.....

.....

.....

.....

.....

.....

Question 18 (4 marks)

The ages of the 10 000 members of the Oztown Sports League are normally distributed with mean 35 years and standard deviation 5 years.

- (a) Use the Empirical Rules to find how many more members are aged in their 30's than are aged in their 40's.

2

- (b) Use the Empirical Rules to find, correct to the nearest whole number, the percentage of the members aged 40 or over who are also aged 50 or over.

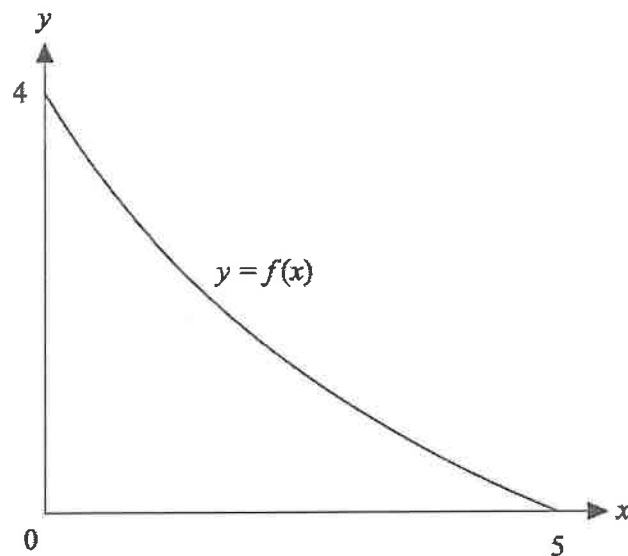
2

Marks**Question 19** (4 marks)

- (a) Describe in correct order a dilation and a translation that can be used to transform the graph of the curve $y = f(x)$ into the graph of the curve $y = f(2x - 1)$.

2

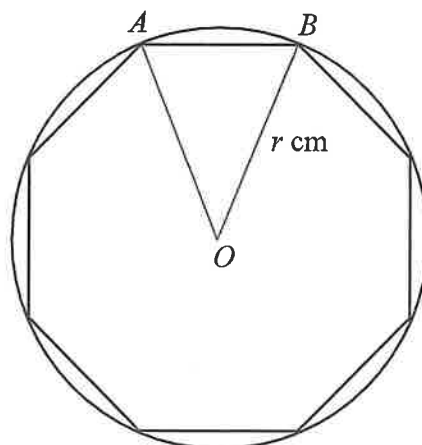
(b)



The diagram shows the graph of the curve $y = f(x)$ with endpoints $(0, 4)$ and $(5, 0)$.

2

On the diagram above, sketch the graph of the curve $y = f(2x - 1)$, showing clearly the coordinates of the endpoints.

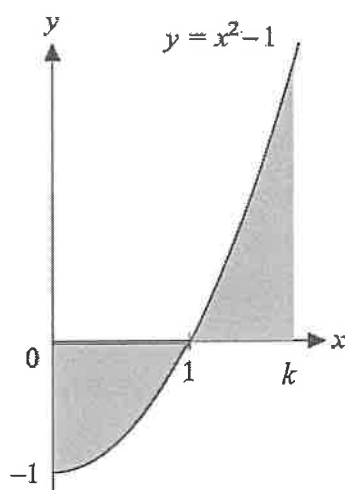
Question 20 (4 marks)

In the diagram $\triangle AOB$ is a section of a regular octagon inscribed in a circle with centre O and radius r cm. The area of the octagon is 80 cm^2 . Find in cm correct to 3 significant figures the perimeter of the octagon.

4**Question 21** (3 marks)

Find in simplest general form the equation of the tangent to the curve $y = \left(x + \frac{2}{x}\right)^2$ at the point where $x = 2$.

3

Question 22 (3 marks)

The diagram shows the graph of the curve $y = x^2 - 1$ for $0 \leq x \leq k$ where $k > 1$. The areas of the two shaded regions are equal. Find in simplest exact form the value of k .

3

Question 23 (4 marks)

The temperature $T^\circ\text{C}$ during the 24 hours from midnight to midnight one day is given by the rule $T = 18 - 12\cos\left(\frac{\pi}{12}(t-3)\right)$ at t hours after midnight for $0 \leq t \leq 24$.

(a) Find the time that day when the temperature reaches its maximum of 30°C .

1

(b) Find correct to the nearest minute the times that day when the temperature is above 20°C .

3

Question 24 (4 marks)

The numbers $4e^{-k}$, 5, $4e^k$ are consecutive terms in an arithmetic sequence.

(a) Show that $2e^{2k} - 5e^k + 2 = 0$.

2

(b) Hence find in simplest exact form any such values of k .

2

Question 25 (4 marks)

(a) Show that the function $f(x) = \frac{\sin x}{1 + \cos x}$ is increasing for $0 < x < \pi$.

2

(b) Find in simplest exact form $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\sin x}{1 + \cos x} dx$.

2

Question 26 (4 marks)

The table below shows the future values of an annuity of \$1 for different rates of interest and for different numbers of compounding periods. The contributions are made at the end of each time period.

<i>Time Period</i>	<i>Future Value Interest Factors</i>				
	<i>Interest Rate</i>				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

- (a) Ozzie wants to open an annuity account in order to have saved \$11 000 in 3 years' time. Equal contributions are to be made at the end of each half-year for the 3 years at an interest rate of 4% p.a. compounding half-yearly. Find the amount of each half-yearly contribution.

2

.....

.....

.....

.....

.....

.....

- (b) Ozzie can only afford to make equal contributions of \$1600 at the end of each half-year for the 3 years. Find the amount by which he falls short of his goal of \$11 000.

2

.....

.....

.....

.....

.....

.....

Question 27 (4 marks)

The mark x in an English test and the mark y in a Mathematics test were recorded for each of a sample of 20 students in a year group. It was found that the least squares regression line of Mathematics marks on English marks had equation $y = 0.64x + 31.68$ and the least squares regression line of English marks on Mathematics marks had equation $x = 0.81y + c$ for some constant c . The mean English mark was $\bar{x} = 63$.

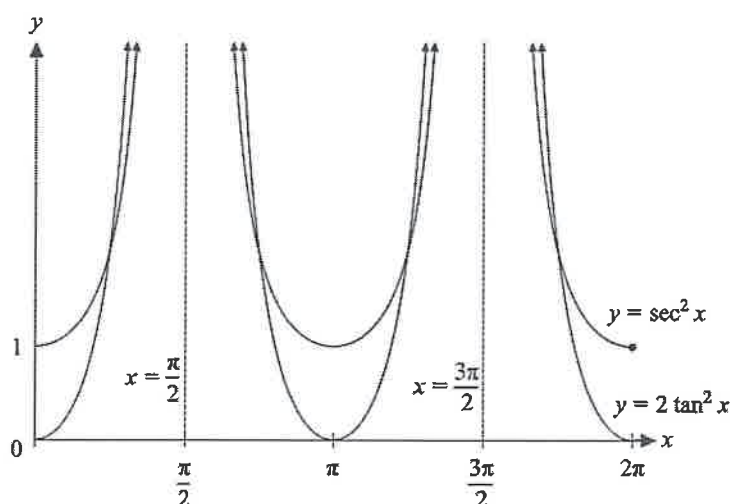
- (a) Find the mean Mathematics mark $\bar{y}$ and the value of the constant c .

2

- (b) An additional student sat the English test but missed the Mathematics test. She found that the estimate of her Mathematics mark using the least squares regression line was the same as her English mark. Find her English mark.

2

Question 29 (6 marks)



The diagram shows the graphs of the curves $y = \sec^2 x$ and $y = 2 \tan^2 x$ for $0 \leq x \leq 2\pi$.

(a) State the periods of each of the two curves.

1

(b) Show that the x coordinates of the points of intersection of the two curves are solutions of the equation $\tan^2 x = 1$ and solve this equation for $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$.

2

(c) Find in simplest exact form the area of the region enclosed by the two curves for $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$.

3

Question 30 (6 marks)

A continuous random variable X has probability distribution function $F(x)$ given by

$$F(x) = \begin{cases} 0, & x < 0 \\ k(20 - 3x)x\sqrt{x}, & 0 \leq x \leq 4 \\ 1, & x > 4 \end{cases} \quad \text{for some constant } k.$$

(a) Show that $k = \frac{1}{64}$.

1

(b) Show that the median of X lies between 1.6 and 1.7.

2

(c) Find in simplest exact form the mode of X .

3

Question 31 (7 marks)

An open topped fish tank of volume 80 m^3 is to be made in the shape of a rectangular prism with length $2x$ metres, width x metres and height h metres. Materials cost \$15 per square metre for the base of the tank and \$16 per square metre for the sides of the tank.

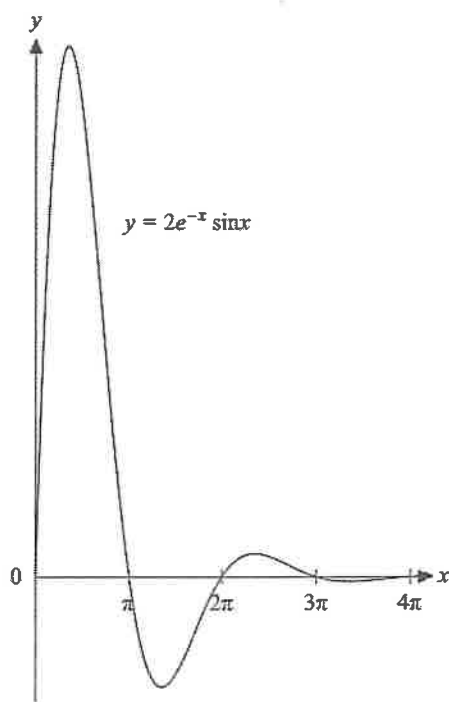
- (a) Show that the total cost \$ C of making the fish tank is given by $C = 30x^2 + \frac{3840}{x}$.

3

- (b) Hence find the dimensions of such a tank with the least total cost.

4

Question 32 (7 marks)



The diagram shows the graph of the curve $y = 2e^{-x}\sin x$ for $x \geq 0$. The successive areas enclosed between the curve and the x axis are denoted by $A_1, A_2, A_3, A_4, \dots$ respectively.

(a) Show that $\frac{d}{dx} \{-e^{-x}(\cos x + \sin x)\} = 2e^{-x}\sin x$.

2

QUESTION 32 CONTINUES ON NEXT PAGE

STUDENT NAME/NUMBER.....

Marks

Section II Extra writing space.

If you use this space, clearly indicate which question you are answering.

[illegible]

STUDENT NAME/NUMBER.....

MarksThis image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

Section 1 Questions 1-10 (1 mark each)

Question	Answer	Solution	Outcomes
1	C	$f(x^2)=0 \quad x^2(x^2-1)=0 \quad x=0, \pm 1$	MA11-2
2	D	$\angle A$ is obtuse with related acute angle opposite side 4 in (3, 4, 5) right triangle. Hence $\sin A = \frac{4}{5}$.	MA11-3
3	A	$f(g(x)) = \log_e(x^2+1) \quad \therefore f(g(x)) \geq 0 \quad \text{Range is } [0, \infty)$	MA11-2
4	B	$P(\bar{A} \cap B) = P(\bar{A})P(B) \quad \therefore 0.4 = P(\bar{A}) \times 0.5 \quad P(\bar{A}) = 0.8 \quad P(A) = 0.2$	MA11-7
5	D	$y = (x^2+4)^{\frac{1}{2}} \quad \frac{dy}{dx} = \frac{1}{2}(x^2+4)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{x^2+4}}$	MA11-5
6	A	$x \rightarrow x + \log_e 2 \rightarrow \frac{x}{2} + \log_e 2 \quad \therefore e^x \rightarrow e^{\frac{x}{2} + \log_e 2} = e^{\frac{x}{2}} e^{\log_e 2} = 2e^{\frac{x}{2}}$	MA12-1
7	B	$z \text{ score is } \frac{70-50}{15} = \frac{4}{3} \quad \text{New mark is } 60 + \frac{4}{3} \times 12 = 76.$	MA12-8
8	C	$\int \frac{2x-1}{2x+1} dx = \int \frac{2x+1-2}{2x+1} dx = \int \left(1 - \frac{2}{2x+1}\right) dx = x - \log_e(2x+1) + c$	MA12-7
9	C	$4000 \left\{ \left(1 + \frac{r}{200}\right)^2 - \left(1 + \frac{r}{100}\right) \right\} = 4000 \left\{ \left(\frac{r}{200}\right)^2 \right\} = \frac{r^2}{10}$	MA12-2
10	D	$\frac{a}{1-r} = \frac{\log_e 2}{1 + \log_e 2} = \frac{\log_e 2}{\log_e e + \log_e 2} = \frac{\log_e 2}{\log_e 2e} = \log_{2e} 2 \quad (\text{Change of base rule})$	MA12-4

Section II

Question 11

Outcomes assessed: MA11-2

Marking Guidelines

Criteria	Marks
Solves both linear equations to obtain two values of x	2
Substantial progress eg. finds one value of x	1

Answer

$$\begin{aligned}
 |2x-1| &= 3 & 2x-1 &= 3 \quad \text{or} \quad 2x-1 = -3 \\
 2x &= 4 & 2x &= -2 \\
 x &= 2 \quad \text{or} \quad x &= -1
 \end{aligned}$$

Question 12

Outcomes assessed: MA11-2

Marking Guidelines

Criteria	Marks
Finds equation of required line	2
Substantial progress eg. finds the gradient of the given line	1

Answer

$$\frac{x}{2} + \frac{y}{3} = 1 \quad y = 3\left(1 - \frac{x}{2}\right)$$

Gradient of given line is $-\frac{3}{2}$

Required $\perp$ line has equation $y = \frac{2}{3}x$

Question 13

a. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Uses circle properties and trigonometry to show required result	2
Substantial progress eg. finds cosine of one of $\angle AOM$, $\angle BOM$	1

Answer

$OA = OB = OP = 12$ (radii of circle) and $OM = 6$ (given $OM = MP$ and $AB \perp OP$)

Hence in right Δ 's AOM and BOM , $\cos \angle AOM = \cos \angle BOM = \frac{6}{12} = \frac{1}{2}$

$$\therefore \angle AOB = \angle AOM + \angle BOM = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

b. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Finds area of sector in simplest exact form	1

Answer

$$\text{Area of sector } AOB \text{ is } \frac{1}{2} \times 12^2 \times \frac{2\pi}{3} = 48\pi \text{ cm}^2$$

Question 14

a. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Uses probability relationships to deduce required result	2
Substantial progress eg. writes $P(A \cap B)$ in terms of $P(A B)$ and $P(B)$	1

Answer

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A|B)P(B) \quad \therefore P(A \cup B) = 0.3 + 0.2 - 0.5 \times 0.2 = 0.4$$

Q14 cont.

b. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Identifies complementary events to state required probability	1

Answer

$$P(\text{neither } A \text{ nor } B) = 1 - P(A \cup B) = 0.6$$

Question 15

Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Uses log laws to write and solve quadratic in x subject to restriction from log function domain	3
Substantial progress eg. correct procedure but quotes 2 solutions, ignoring restriction on x	2
Some progress eg. uses log laws to obtain quadratic equation in x	1

Answer

$$2\log_2(x-6) = \log_2 x$$

$$\log_2(x-6)^2 = \log_2 x, \quad x > 6$$

$$(x-6)^2 = x, \quad x > 6$$

$$x^2 - 13x + 36 = 0$$

$$(x-9)(x-4) = 0$$

$$\therefore x = 9$$

Question 16

Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Uses log laws and appropriate trig identities to prove required result	3
Substantial progress eg. correct procedure but some lack of clarity in explanation	2
Some progress eg. uses appropriate log law and some trig identities to begin process	1

Answer

$$\begin{aligned}
 \log_e(1 + \sin x) - \log_e(1 - \sin x) &= \log_e\left(\frac{1 + \sin x}{1 - \sin x}\right) \\
 &= \log_e\left(\frac{(1 + \sin x)^2}{1 - \sin^2 x}\right) \\
 &= \log_e\left(\frac{(1 + \sin x)^2}{\cos^2 x}\right) \\
 &= 2\log_e\left(\frac{1 + \sin x}{\cos x}\right) \\
 &= 2\log_e(\sec x + \tan x)
 \end{aligned}$$

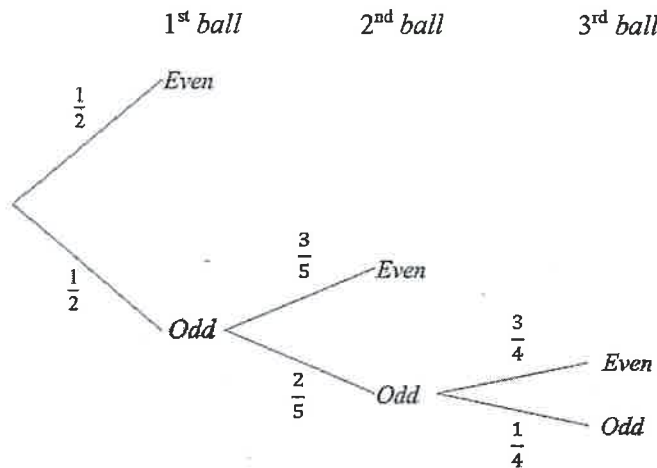
Question 17

a. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Completes probability tree	1

Answer



b. Outcomes assessed:

Marking Guidelines

Criteria	Marks
Completes probability distribution table and states probability of specified event	3
Substantial progress eg. completes table but does not state correct probability	2
Some progress eg. writes 2 or 3 correct entries in table	1

Answer

x	0	1	2	3
$p(x)$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{3}{20}$	$\frac{1}{20}$

For a majority of odd balls need *odd, odd, even* or *odd, odd, odd* i.e. $X=2$ or $X=3$

$$\therefore P(\text{majority odd}) = \frac{3}{20} + \frac{1}{20} = \frac{1}{5}$$

Question 18

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses appropriate ranges of z scores and the Empirical Rules to get the required estimate	2
Substantial progress eg. estimates the number of members in their 30's	1

Answer

Members in their 30's have $-1 \leq z < 1$, giving 68% of 10000 = 6800 members.

Members in their 40's have $1 \leq z < 3$, giving $\frac{1}{2}(99.7 - 68)\%$ of 10000 = 1585 members.

Hence there are 5215 more members aged in their 30's than in their 40's.

Q18 cont.

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses appropriate ranges of z scores and the Empirical Rules to get the required estimate	2
Substantial progress eg. estimates the number of members aged over 40	1

Answer

Members aged 40 or over have $z \geq 1$, giving $\frac{1}{2}(100 - 68)\%$ of $10\,000 = 1600$ members
 $1600 - 1585 = 15$ of these are aged 50 or over. Hence the % of members aged 40 or over who are aged 50 or over is $\left(\frac{15}{1600} \times 100\right)\% \approx 1\%$

Question 19

a. Outcomes assessed: MA12-1

Marking Guidelines

Criteria	Marks
Describes an appropriate dilation and translation in the correct order.	2
Substantial progress eg. describes one transformation correctly given the order presented.	1

Answer

$x \rightarrow 2x \rightarrow 2\left(x - \frac{1}{2}\right)$, corresponding to a horizontal dilation by a factor $\frac{1}{2}$ followed by a horizontal translation in the positive x direction by $\frac{1}{2}$ a unit.

Alternatively,

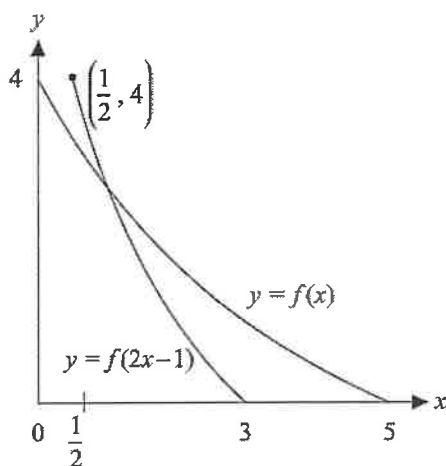
$x \rightarrow x - 1 \rightarrow 2x - 1$, corresponding to a horizontal translation in the positive x direction by 1 unit followed by a horizontal dilation by a factor $\frac{1}{2}$.

b. Outcomes assessed: MA12-1

Marking Guidelines

Criteria	Marks
Sketches graph showing required information	2
Substantial progress eg. curve of correct shape with only one of the endpoints correct	1

Answer



Question 20

Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Finds the perimeter of the octagon correct to 3 sig. fig.	4
Substantial progress eg. correct procedure but an error in calculation or accuracy	3
Moderate progress eg. finds r^2 exactly and an expression for AB or AB^2 using trigonometry	2
Some progress eg. finds the exact value of r^2	1

Answer

$$\text{Area } \triangle AOB = 10 \text{ cm}^2. \therefore \frac{1}{2}r^2 \sin \frac{2\pi}{8} = 10 \quad \therefore r^2 = 20\sqrt{2}$$

$$\text{Using the cosine rule in } \triangle AOB, AB^2 = r^2 + r^2 - 2r^2 \cos \frac{2\pi}{8} = 2r^2 \left(1 - \cos \frac{\pi}{4}\right) = 40\sqrt{2} \left(1 - \frac{1}{\sqrt{2}}\right)$$

Hence the perimeter of the octagon is $8\sqrt{40(\sqrt{2}-1)} \approx 32.6 \text{ cm}$ (3 sig. fig.)

Question 21

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Uses differentiation to find gradient and then equation of required tangent	3
Substantial progress eg. differentiates to find gradient of the tangent when $x=2$	2
Some progress eg. applies chain rule correctly	1

Answer

$$y = \left(x + \frac{2}{x}\right)^2 \quad \frac{dy}{dx} = 2\left(x + \frac{2}{x}\right)\left(1 - \frac{2}{x^2}\right)$$

$$x=2 \Rightarrow y=9 \text{ and } \frac{dy}{dx} = 3$$

Tangent at (2, 9) has gradient 3 and equation

$$y - 9 = 3(x - 2) \quad 3x - y + 3 = 0$$

Question 22

Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Uses integration to find the value of k	3
Substantial progress eg. integrates to find a cubic equation for k	2
Some progress eg. writes an equation involving definite integral(s) and finds primitive function	1

Answer

Given the shaded areas are equal, $\int_0^k (x^2 - 1) dx = 0$.

$$\left[\frac{1}{3}x^3 - x\right]_0^k = 0$$

$$\frac{1}{3}k^3 - k = 0$$

$$\frac{1}{3}k(k^2 - 3) = 0$$

But $k > 1$ hence $k = \sqrt{3}$

Question 23

a. Outcomes assessed: MA12-5

Marking Guidelines

Criteria	Marks
Finds time when temperature is a maximum	1

Answer

Maximum T occurs when $\cos\left(\frac{\pi}{12}(t-3)\right) = -1$, $0 \leq t \leq 24$ $\therefore \frac{1}{12}(t-3) = 1$ $\therefore t = 15$

Temperature is a maximum at 3 pm.

b. Outcomes assessed: MA12-5

Marking Guidelines

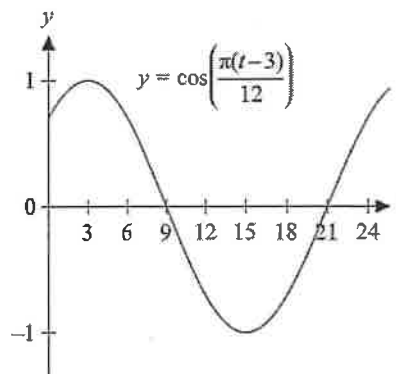
Criteria	Marks
Finds the required times correct to the nearest minute	3
Substantial progress eg. finds one t value for $T=20$ and converts to time to the nearest minute	2
Some progress eg. finds expression involving t and calculated value of $\cos^{-1}\left(-\frac{1}{6}\right)$	1

Answer

$$18 - 12\cos\left(\frac{\pi}{12}(t-3)\right) > 20$$

$$12\cos\left(\frac{\pi}{12}(t-3)\right) < -2$$

$$\cos\left(\frac{\pi}{12}(t-3)\right) < -\frac{1}{6}$$



Considering the graph (dilation horizontally by factor 12 and horizontal translation 3 units to right of graph of $y = \cos \pi t$), we require t to lie between the two solutions of

$$\begin{aligned} \cos\left(\frac{\pi}{12}(t-3)\right) &= -\frac{1}{6} & \frac{\pi}{12}(t-3) &= \pi - \cos^{-1}\left(\frac{1}{6}\right), \pi + \cos^{-1}\left(\frac{1}{6}\right) \\ t &= \frac{12}{\pi}\left(\pi \mp \cos^{-1}\left(\frac{1}{6}\right)\right) + 3, \\ t &\approx 9.6396, \quad 20.3604 \end{aligned}$$

Hence temperature exceeds 20°C between 9.38 am and 8.22 pm.

$$\text{Alternatively, calculator gives } \frac{\pi}{12}(t-3) = \cos^{-1}\left(-\frac{1}{6}\right) \approx 1.7382 \quad t \approx 9.6396$$

so earliest time $T = 20$ is 9.38 am. Using symmetry in the graph, latest time $T = 20$ will be 38 minutes before 9 pm, i.e. at 8.22 pm.

Question 24

a. Outcomes assessed: MA12-4

Marking Guidelines

Criteria	Marks
Uses defining property of A.P. to write and simplify equation for k into required form	2
Substantial progress eg. uses defining property of A.P. to write equation for k	1

Answer

$$5 - 4e^{-k} = 4e^k - 5$$

$$4e^k - 10 + 4e^{-k} = 0$$

$$4e^k e^k - 10e^k + 4 = 0$$

$$2e^{2k} - 5e^k + 2 = 0$$

b. Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Solves for k in simplest exact form	2
Substantial progress eg. finds values of e^k by solving an appropriate quadratic equation	1

Answer

$$2(e^k)^2 - 5e^k + 2 = 0$$

$$(2e^k - 1)(e^k - 2) = 0$$

$$e^k = \frac{1}{2} \quad \text{or} \quad e^k = 2$$

$$k = \ln\left(\frac{1}{2}\right) = -\ln 2 \quad \text{or} \quad k = \ln 2$$

Question 25

a. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Considers sign of first derivative to deduce result	2
Substantial progress eg. finds first derivative	1

Answer

$$f(x) = \frac{\sin x}{1 + \cos x} \quad f'(x) = \frac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2}$$

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$$

$$= \frac{\cos x + 1}{(1 + \cos x)^2}$$

$$= \frac{1}{1 + \cos x}$$

For $0 < x < \pi$, $-1 < \cos x < 1$ $\therefore 1 + \cos x > 0$ and $f'(x) > 0$. $\therefore f(x)$ is increasing for $0 < x < \pi$.

Q25 cont.

b. Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Evaluates definite integral in simplest exact form	2
Substantial progress eg. finds anti-derivative	1

Answer

$$\begin{aligned}
 \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\sin x}{1 + \cos x} dx &= - \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{-\sin x}{1 + \cos x} dx \\
 &= - \left[\ln(1 + \cos x) \right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \\
 &= - \left(\ln \frac{1}{2} - \ln \frac{3}{2} \right) \\
 &= \ln 3
 \end{aligned}$$

Question 26

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Uses table to calculate instalment	2
Substantial progress eg. selects appropriate table entry	1

Answer

2% per half-year over 6 half-years. $\therefore$ Instalment \$ a is given by $6 \cdot 3081 \times a = 11\,000$ $a \approx 1743.79$
Half-yearly instalment is \$1743.79.

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Uses table to calculate shortfall	2
Substantial progress eg. correct process but only one of rate or number of periods correct	1

Answer

Shortfall is $\$(11\,000 - 1600 \times 6 \cdot 3081) = \907.04

Question 27

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses regression line of y on x to calculate $\bar{y}$ then the regression line of x on y to find c	2
Substantial progress eg. finds $\bar{y}$	1

Answer

Both least squares regression lines go through $(\bar{x}, \bar{y})$.

$$\therefore \bar{y} = 0.64\bar{x} + 31.68 = 0.64 \times 63 + 31.68 \quad \bar{y} = 72$$

Then $\bar{x} = 0.81\bar{y} + c$ gives $c = 63 - 0.81 \times 72 = 4.68$

Q27 cont.

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses the appropriate regression line to calculate x	2
Substantial progress eg. correct process but uses wrong regression line	1

Answer

$$y = 0.64x + 31.68 \text{ and } y = x \quad \therefore x - 0.64x = 31.68 \quad \therefore \text{English mark is } x = \frac{31.68}{0.36} = 88$$

Question 28

a. Outcomes assessed: MA12-4

Marking Guidelines

Criteria	Marks
Applies the formula for sum of a G.P and simplifies to obtain required expression	2
Substantial progress eg. applies the formula for sum of a G.P	1

Answer

$$\begin{aligned} P_n &= 1000(1.1)^n - Q \left(\frac{(1.1)^n - 1}{1.1 - 1} \right) \\ &= 1000(1.1)^n - 10Q((1.1)^n - 1) \\ &= (1.1)^n(1000 - 10Q) + 10Q \end{aligned}$$

b. Outcomes assessed: MA12-4

Marking Guidelines

Criteria	Marks
Finds an expression for P_{20+n} in terms of n , solves $P_{20+n} = 1000$, states required calendar year.	4
Substantial progress eg. correct procedure with one error or does not interpret n value	3
Moderate progress eg. writes and simplifies P_{20+n} in terms of n	2
Some progress eg. writes P_{20+n} in terms of P_{20}	1

Answer

$$\begin{aligned} P_{20} &= (1.1)^{20}(1000 - 800) + 800 \\ P_{20+n} &= (1.1)^n(P_{20} - 3000) + 3000 \\ &= (1.1)^n\{200(1.1)^{20} + 800 - 3000\} + 3000 \\ &= (1.1)^n\{200(1.1)^{20} - 2200\} + 3000 \\ P_{20+n} &= 1000 \Rightarrow (1.1)^n\{200(1.1)^{20} - 2200\} = -2000 \end{aligned}$$

$$(1.1)^n\{11 - (1.1)^{20}\} = 10$$

$$(1.1)^n = \frac{10}{11 - (1.1)^{20}}$$

$$n \ln 1.1 = \ln \left(\frac{10}{11 - (1.1)^{20}} \right)$$

$$n = \frac{\ln 10 - \ln\{11 - (1.1)^{20}\}}{\ln(1.1)}$$

$$n \approx 8.9$$

Q28 cont.

At the end of 2053, after the 300 deer have been removed, population is $P_{20+9} \approx 985$, and this is the first time the population has dropped below 1000.

Question 29**a. Outcomes assessed: MA11-4****Marking Guidelines**

Criteria	Marks
States the common period of the two curves	1

Answer

Curves $y = \sec^2 x$ and $y = 2\tan^2 x$ each have period π .

b. Outcomes assessed: MA11-3**Marking Guidelines**

Criteria	Marks
Establishes the required equation and solves for x in the stated domain	2
Substantial progress eg. establishes the required equation	1

Answer

Curves intersect where

$$\begin{aligned} 2\tan^2 x &= \sec^2 x \\ &= 1 + \tan^2 x \\ \tan^2 x &= 1 \end{aligned}$$

$$\tan x = \pm 1$$

$$x = \pi \pm \frac{\pi}{4} \quad \text{for } \frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$$

$$\therefore x = \frac{3\pi}{4} \quad \text{or} \quad x = \frac{5\pi}{4}$$

c. Outcomes assessed: MA12-7**Marking Guidelines**

Criteria	Marks
Finds the required area by evaluating an appropriate definite integral	3
Substantial progress eg. finds the primitive of an appropriate definite integral	2
Some progress eg. writes an appropriate definite integral in terms of $\sec^2 x$	1

Answer

$$\begin{aligned} \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} (\sec^2 x - 2\tan^2 x) dx &= \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} \{ \sec^2 x - 2(\sec^2 x - 1) \} dx \\ &= \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} (2 - \sec^2 x) dx \\ &= \left[2x - \tan x \right]_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} \\ &= 2\left(\frac{5\pi}{4} - \frac{3\pi}{4}\right) - \left\{ \tan\left(\frac{5\pi}{4}\right) - \tan\left(\frac{3\pi}{4}\right) \right\} \\ &= \pi - 2 \end{aligned}$$

Hence required area between the curves is $(\pi - 2)$ sq. units.

Question 30

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Establishes the required value of k	1

Answer

Clearly $F(0) = 0$. Also require $F(4) = 1$. $\therefore k \times (20 - 12) \times 4 \times 2 = 1$
 $64k = 1 \quad \therefore k = \frac{1}{64}$

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Establishes the required result	2
Substantial progress eg. evaluates one of $F(1.6)$ or $F(1.7)$ correctly	1

Answer

The median of X is given by $F(x) = \frac{1}{2}$. $F(x)$ is a continuous, increasing function such that
 $F(1.6) = \frac{1}{64}(20 - 4 \cdot 8) \times 1.6 \times \sqrt{1.6} \approx 0.481 < \frac{1}{2}$ and $F(1.7) = \frac{1}{64}(20 - 5 \cdot 1) \times 1.7 \times \sqrt{1.7} \approx 0.516 > \frac{1}{2}$.
 $\therefore F(x) = \frac{1}{2}$ for some x such that $1.6 < x < 1.7$. Hence the median of X lies between 1.6 and 1.7.

c. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Finds the value of x that maximises the probability density function $f(x) = F'(x)$	3
Substantial progress eg. finds $f'(x)$ by finding $F''(x)$	2
Some progress eg. finds $f(x) = F'(x)$	1

Answer

The mode of X is the value of x for which $f(x) = F'(x)$ takes its maximum value.

$$\begin{aligned}
 f(x) &= \frac{d}{dx} \frac{1}{64} (20 - 3x)x^{\frac{3}{2}} & f'(x) &= \frac{15}{128} \left(2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} \right) & f'(x) &= 0 \\
 &= \frac{1}{64} \left\{ -3x^{\frac{3}{2}} + (20 - 3x)\frac{3}{2}x^{\frac{1}{2}} \right\} & &= \frac{45}{256} \frac{1}{\sqrt{x}} \left(\frac{4}{3} - x \right) & x &= \frac{4}{3} \\
 &= \frac{1}{64} \left\{ 30x^{\frac{1}{2}} - \frac{15}{2}x^{\frac{3}{2}} \right\} & f''(x) &= \frac{15}{128} \left(-x^{-\frac{3}{2}} - \frac{3}{4}x^{-\frac{1}{2}} \right) & f''\left(\frac{4}{3}\right) &< 0 \\
 &= \frac{15}{128} \left\{ 4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right\} & &< 0 \text{ for } 0 < x \leq 4
 \end{aligned}$$

Hence $f(x)$ takes its maximum value for $x = \frac{4}{3}$. $\therefore$ the mode of X is $\frac{4}{3}$.

Question 31

a. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Establishes required expression for C in terms of x	3
Substantial progress eg. finds h in terms of x and C in terms of x and h	2
Some progress eg. finds only one of h in terms of x , or C in terms of x and h	1

Answer

$$\begin{aligned}
 V &= 80 & C &= 2x^2 \times 15 + 2(2xh + xh) \times 16 \\
 2x^2h &= 80 & &= 30x^2 + 96xh \\
 h &= \frac{40}{x^2} & &= 30x^2 + 96x \left(\frac{40}{x^2} \right) \\
 & & &\therefore C = 30x^2 + \frac{3840}{x}
 \end{aligned}$$

b. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Uses differentiation to find value of x that minimises C and corresponding tank dimensions	4
Substantial progress eg. correct procedure except that value of C not verified as minimum	3
Moderate progress eg. finds x for min C , verifying minimum, but 3 dimensions not stated	2
Some progress eg. differentiates C with respect to x	1

Answer

$$\begin{aligned}
 C &= 30x^2 + \frac{3840}{x} & \frac{dC}{dx} &= 0 \quad \text{and } x > 0 \\
 \frac{dC}{dx} &= 60x - \frac{3840}{x^2} & \frac{d^2C}{dx^2} &= 60 - 3840(-2x^{-3}) \\
 &= \frac{60}{x^2}(x^3 - 64) & &= 60 \left(1 + \frac{128}{x^3} \right) & x &= 4 \\
 & & & & \frac{d^2C}{dx^2} &> 0
 \end{aligned}$$

Hence C takes its minimum value for $x = 4$ and $h = \frac{40}{16} = 2.5$

Hence minimum total cost for tank with base 8m by 4m and height 2.5 m.

Question 32

a. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Applies the product rule and simplifies into required form	2
Substantial progress eg. applies the product rule	1

Answer

$$\begin{aligned}
 \frac{d}{dx} \{ -e^{-x}(\cos x + \sin x) \} &= e^{-x}(\cos x + \sin x) - e^{-x}(-\sin x + \cos x) \\
 &= e^{-x}(\cos x + \sin x + \sin x - \cos x) \\
 &= 2e^{-x} \sin x
 \end{aligned}$$

Q32 cont.

b. Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Establishes the two required results by evaluating appropriate definite integrals	2
Substantial progress eg. establishes one of the two required results	1

Answer

$$\begin{aligned}
 A_1 &= \int_0^{\pi} 2e^{-x} \sin x dx \\
 &= \left[-e^{-x}(\cos x + \sin x) \right]_0^{\pi} \\
 &= -e^{-\pi}(-1+0) - (-1)(1+0) \\
 &= 1 + e^{-\pi}
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= - \int_{\pi}^{2\pi} 2e^{-x} \sin x dx \\
 &= \left[e^{-x}(\cos x + \sin x) \right]_{\pi}^{2\pi} \\
 &= e^{-2\pi}(1+0) - e^{-\pi}(-1+0) \\
 &= e^{-\pi}(1 + e^{-\pi})
 \end{aligned}$$

c. Outcomes assessed: MA12-4

Marking Guidelines

Criteria	Marks
Solves $A_n < 10^{-9}$ for integer values of n	3
Substantial progress eg. obtains an inequality for n or $(n-1)$	2
Some progress eg. writes an expression for A_n in terms of n	1

Answer

For this G.P., $a = (1 + e^{-\pi})$, $r = A_2 \div A_1 = e^{-\pi}$.

$$\begin{aligned}
 A_n &= (1 + e^{-\pi})(e^{-\pi})^{n-1} \\
 &= e^{-(n-1)\pi}(1 + e^{-\pi})
 \end{aligned}$$

$$A_n < 10^{-9} \quad \text{for} \quad e^{-(n-1)\pi}(1 + e^{-\pi}) < 10^{-9}$$

$$e^{-(n-1)\pi} < \frac{10^{-9}}{(1 + e^{-\pi})}$$

$$e^{(n-1)\pi} > \frac{(1 + e^{-\pi})}{10^{-9}}$$

$$(n-1)\pi > \ln(1 + e^{-\pi}) + 9\ln 10$$

$$n > \frac{\ln(1 + e^{-\pi}) + 9\ln 10}{\pi} + 1 \approx 7.61$$

$$\therefore n = 8, 9, 10, \dots$$

Question	Marks	Content	Syllabus Outcomes	Targeted Performance Bands
1	1	Working with functions	MA11-2	2-3
2	1	Trigonometry and measure of angles	MA11-3	2-3
3	1	Working with functions	MA11-2	3-4
4	1	Probability and discrete probability distributions	MA11-7	3-4
5	1	Introduction to differentiation	MA11-5	3-4
6	1	Graphing techniques	MA12-1	4-5
7	1	Random variables	MA12-8	4-5
8	1	Integral calculus	MA12-7	4-5
9	1	Modelling financial situations	MA12-2	5-6
10	1	Modelling financial situations	MA12-4	5-6
11	2	Working with functions	MA11-2	2-3
12	2	Working with functions	MA11-2	2-3
13 a	2	Trigonometry and measure of angles	MA11-3	2-3
b	1	Trigonometry and measure of angles	MA11-3	2-3
14 a	2	Probability and discrete probability distributions	MA11-7	2-3
b	1	Probability and discrete probability distributions	MA11-7	2-3
15	3	Logarithms and exponentials	MA11-6	3-4
16	3	Trigonometric functions and identities	MA11-3	4-5
17 a	1	Probability and discrete probability distributions	MA11-7	2-3
b	3	Probability and discrete probability distributions	MA11-7	3-4
18 a	2	Random variables	MA12-8	2-3
b	2	Random variables	MA12-8	3-4
19 a	2	Graphing techniques	MA12-1	2-3
b	2	Graphing techniques	MA12-1	3-4
20	4	Trigonometry and measure of angles	MA11-3	5-6
21	3	Differential calculus	MA12-6	3-4
22	3	Integral calculus	MA12-7	3-4
23 a	1	Trigonometric functions and graphs	MA12-5	4-5
b	3	Trigonometric functions and graphs	MA12-5	4-5
24 a	2	Modelling financial situations	MA12-4	4-5
b	2	Logarithms and exponentials	MA11-6	3-4
25 a	2	Applications of differentiation	MA12-6	3-4
b	2	Integral calculus	MA12-7	4-5
26 a	2	Modelling financial situations	MA12-2	2-3
b	2	Modelling financial situations	MA12-2	2-3
27 a	2	Descriptive statistics and bivariate data analysis	MA12-8	3-4
b	2	Descriptive statistics and bivariate data analysis	MA12-8	4-5
28 a	2	Modelling financial situations	MA12-4	4-5
b	4	Modelling financial situations	MA12-4	5-6
29 a	1	Trigonometric functions and identities	MA11-4	2-3
b	2	Trigonometric functions and identities	MA11-3	4-5
c	3	Integral calculus	MA12-7	5-6
30 a	1	Random variables	MA12-8	2-3
b	2	Random variables	MA12-8	2-3
c	3	Random variables	MA12-8	5-6
31 a	3	Working with functions	MA11-2	5-6
b	4	Applications of differentiation	MA12-6	5-6

32 a	2	Applications of differentiation	MA12-6	5-6
b	2	Integral calculus	MA12-7	4-5
c	3	Modelling financial situations	MA12-4	4-5